

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Third Semester of B. Tech. Theory Examination (CE / IT)****March -Apr 2018****MA242 Discrete Mathematics****Date: 26.03.2018, Monday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

- i. The question paper contains two sections.
- ii. Section I and II must be attempted in separate answer sheets.
- iii. Figures to the right indicate marks.
- iv. Make suitable assumptions and draw neat figure where it is required.
- v. All notations and terminologies are standard.

Section-I**Q-1 Select the correct option for following multiple choice questions. [07]**

- 1) If a set A has n elements, then how many relations can be defined from set A to A? **01**
 a) n^2 b) 2^{n^2} c) 2^n d) n
- 2) Which of the following is true? **01**
 a) $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$ b) $\neg p \Rightarrow q \equiv q \Rightarrow p$
 c) $q \Rightarrow p \equiv p \Rightarrow \neg q$ d) $p \Rightarrow q \equiv \neg p \Rightarrow \neg q$
- 3) Which of the following is always true for the lattice? **01**
 a) Contains unique minimal element. b) Contains unique maximal element.
 c) Every lattice is a POSET. d) Every POSET is a lattice.
- 4) Let $(L, *, \oplus)$ be any lattice and $a, b, c \in L$. If $b \leq a$ then which of the following is true? **01**
 a) $a * b = a$ b) $a * c \leq b * c$
 c) $a \oplus b = b$ d) $b \oplus c \leq a \oplus c$
- 5) Which of the following is true for the multiplicative group $G = \{1, -1, i, -i\}$? **01**
 a) G forms an Abelian multiplicative group. b) G forms a multiplicative group but not an Abelian group.
 c) G does not form a multiplicative group. d) G is not associative under multiplication.
- 6) Which of the following operation is a binary operation? **01**
 a) $x * y := x \div y \quad \forall x, y \in \mathbb{R}$ b) $x * y := x - y \quad \forall x, y \in \mathbb{N}$
 c) $x * y := x - y \quad \forall x, y \in \mathbb{R}$ d) $x * y := x \cdot y \quad \forall x, y \in \mathbb{Q}^c$
- 7) Which of the following is not a group? **01**
 a) $(\mathbb{R}, +)$ b) $(M_n(\mathbb{R}), \cdot)$
 c) $(\mathbb{Z}, +)$ d) $(M_n(\mathbb{R}), +)$

Q-2 Attempt any four out of six . [16]

- 1) Obtain principal conjunctive normal form for the logical expression $q \wedge (p \vee \neg q)$ with help of truth table. **04**
- 2) Check whether the set $\{\neg, \vee\}$ is a minimal functionally complete set of connectives or not. **04**
- 3) Define inverse, converse and contra-positive for the conditional statement $p \Rightarrow q$ and discuss logical equivalence between them. **04**
- 4) Let $(L, *, \oplus)$ be a lattice, then prove that $a \leq b \Leftrightarrow a * b = a \Leftrightarrow a \oplus b = b$ **04**
- 5) State and prove absorption properties for elements of a lattice $(L, *, \oplus)$. **04**
- 6) Use Karnaugh map to simplify the following Boolean function: **04**

$$f(A, B, C) = A'B'C' + A'B'C + A'BC + A'BC' + AB'C + ABC$$

Q-3

- 1) **Attempt any two out of three . [06]**
 - a) Show that there is no Boolean algebra with three elements only. **03**
 - b) Let $R := \{(x, y) : x \in \mathbb{Z}, y \in \mathbb{Z} \text{ and } x - y \text{ is divisible by } 7\}$ be a relation on $\mathbb{Z}$ then show that the relation R is an equivalence relation on $\mathbb{Z}$. **03**
 - c) In any Boolean algebra $(B, \cdot, +, ', 0, 1)$ prove that $a = 0 \Leftrightarrow a \cdot b' + a' \cdot b = b$. **03**
 - 2) Check whether the set $G = \{0, 1, 2, 3, 4, 5\}$ forms an abelian group of order 6 under the binary operation addition modulo 6 or not. **[06]**
- OR**
- 2) Prove that the set of cube roots of unity $G = \{1, \omega, \omega^2\}$ form a cyclic multiplicative group. **[06]**

Section-II

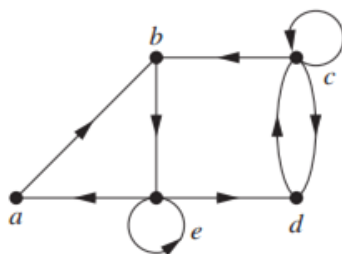
Q-4 Select the correct option for following multiple choice questions. [07]

- 1) How many edges are there in a graph with 12 vertices each of degree four? **01**
 a) 12 b) 48 c) 24 d) 6
- 2) Which of the following is correct for a simple connected planar graph with $v \geq 3$? **01**
 a) $e \geq 3v - 6$ b) $e \leq 3v - 6$ c) $3v \leq e - 6$ d) $3v \geq e - 6$
- 3) A tree with n vertices has _____ edges. **01**
 a) 2n b) n c) n+1 d) n - 1

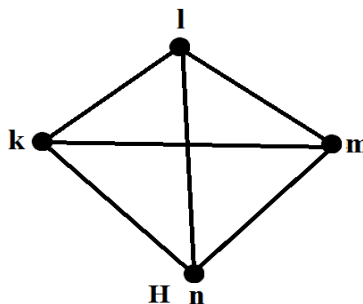
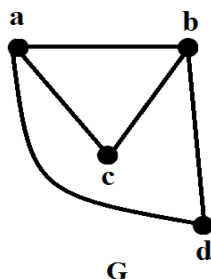
- 4) Sum of characteristic values of the matrix $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ = _____. 01
- a) 10 b) 12 c) 6 d) 30
- 5) The characteristic equation of the matrix $A = \begin{bmatrix} 4 & 0 \\ 5 & -1 \end{bmatrix}$ is equal to _____. 01
- a) $\lambda^2 - 3\lambda - 4 = 0$ b) $\lambda^2 - 3\lambda + 4 = 0$
c) $\lambda^2 + 3\lambda - 4 = 0$ d) $\lambda^2 + 3\lambda + 4 = 0$
- 6) The recurrence relation $a_n = 4na_{n-1} + 3^n$ is _____. 01
- a) Linear and Homogeneous b) Nonlinear and Non homogeneous
c) Nonlinear and Homogeneous d) Linear and Non Homogeneous
- 7) What form does a particular solution of the linear non-homogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$ have when $F(n) = 3^n$? 01
- a) $a_n^{(p)} = p_0 n^2 3^n$ b) $a_n^{(p)} = p_0 3^n$
c) $a_n^{(p)} = n^2 (p_1 n + p_0) 3^n$ d) $a_n^{(p)} = (p_2 n^2 + p_1 n + p_0) 3^n$

Q-5 Attempt any four out of six . [16]

- 1) Determine degree of each vertex in the given graph 04



- 2) Define Graph isomorphism and check whether following graphs are isomorphic or not. 04



- 3) Let T be a tree with 30 edges. The removal of a certain edge from T yields two disjoint trees T_1 and T_2 . Given that the number of vertices in T_1 equals the number of edges in T_2 , determine the number of vertices and edges in T_1 and T_2 . 04

4) Show that the matrix $A = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$ is unitary and find its eigenvalues. 04

5) Find the eigen values and their corresponding eigen vectors of A^{25} where 04

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}.$$

6) If $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$ then show that A is Hermitian matrix and iA is skew - 04

Hermitian matrix.

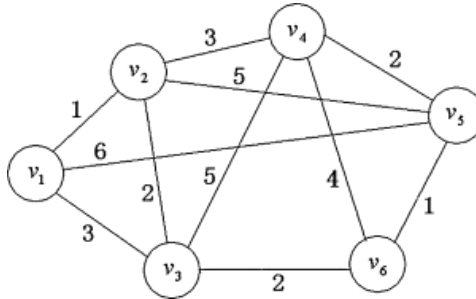
Q-6

1) Attempt any two out of three . [06]

a) Construct the labeled tree for the algebraic expression 03

$$(((x-y)*z)-3) \div (19+(x*x)).$$

b) Use Kruskal's algorithm to find a minimum spanning tree 03



c) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$. 03

2) Solve the recurrence relation $a_r + 3a_{r-1} + 2a_{r-2} = f(r)$; $f(r) = \begin{cases} 1 & r = 2 \\ 0 & \text{otherwise} \end{cases}$, [06]

with the boundary condition $a_0 = a_1 = 0$.

OR

2) Find all solutions of the recurrence relation $a_n = 5a_{n-1} + 6a_{n-2} + 7^n$. [06]